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Evaluation of an improved CFD model against nine vented deflagration experiments

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HIGHLIGHTS

- Evaluation of a newly developed CFD deflagration model.
- Hydrogen vented deflagration experiments of KIT and FM-Global are analyzed.
- Back and central ignition cases with or without initial turbulence are examined.
- Simulations predict turbulence development outside the vent.
- Overall, satisfactory performance of the CFD model.

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ABSTRACT

In the present work, a newly developed CFD deflagration model incorporated into the ADREA-HF code is evaluated against hydrogen vented deflagrations experiments carried out by KIT and FM-Global in a medium (1 m^3) and a real (63.7 m^3) scale enclosure respectively. A square vent of 0.5 m^2 and 5.4 m^2 respectively is located in the center of one of side walls. In the case of the medium scale enclosure the 18% v/v homogeneous hydrogen-air mixture and back-wall ignition case is examined. In the case of the real scale enclosure the examined cases cover different homogeneous mixture concentrations (15% and 18% v/v), different ignition locations (back-wall and center) and different levels of initial turbulence. The CFD model accounts for flame instabilities that develop as the flame propagates inside the chamber and turbulence that mainly develops outside the vent. Pressure predictions are compared against experimental measurements revealing a very good performance of the CFD model for the back-wall ignition cases. For the center ignition cases, the model overestimates the maximum overpressure. The opening of the vent cover is identified as a possible reason for the overprediction. The analysis indicates that turbulence is the main factor which enhances external explosion strength causing the sudden pressure increase, confirming previous findings.

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Introduction

Hydrogen use is expected to increase in the near future and its explosive nature brings up significant safety issues. In the case of an accident, released hydrogen is mixed with air and forms a flammable cloud. An ignition of hydrogen and flame propagation in closed space can have catastrophic consequences. The confined space will lead to the development of much higher over-pressure compared to a similar explosion in open space. The venting of the compartment is the final protection measure in order to mitigate the damages of the explosion. In the past years, the increase of computational power has rendered Computational Fluid Dynamics (CFD) as a very attractive methodology for risk assessment [1]. With its high numerical accuracy, it can evaluate regulations and standards and give deeper insight into the physical phenomenon.

The interest in vented deflagrations has increased recently as it is shown by the large number of publications (e.g. Refs. [2–16]). External explosion is a phenomenon of increased importance in vented geometries because it can significantly increase the internal pressure. As the combustion develops inside the enclosure, unburned mixture is pushed outside through the vent. When the flame exits the vent, the mixture outside starts burning creating a secondary explosion. This explosion seems to have much more strength than the internal one. The main reasons that have been proposed for that behavior are the creation of extensive turbulence in the area outside the vent because of the jet-like flow, and the development of Rayleigh-Taylor instability due to the acceleration of the light products through the vent to the dense unburned mixture. Both reasons potentially lead to an augmentation of the reaction rate which leads to a violent external explosion. The pressure increase in the area outside the vent causes the decrease of the venting rate and consequently the increase of the internal pressure. An analysis of the peak over-pressures in vented explosion was made by Chao et al. [17], where the effects of fuel, enclosure size, ignition location, vent size and obstacles were examined. Special attention to external explosion has been paid in several works such as [2,5,8,18–20].

CFD modeling of deflagrations is a big challenge. Combustion occurs at very small scales of the order of millimeters whereas geometries of practical interest are of the order of meters. The growth of instabilities at the flame front and the complex interaction between the flame and the turbulence make the development of a global model a difficult task. External explosion introduce an additional difficulty in deflagration modelling. Molkov et al. [21,22] performed Large Eddy Simulation (LES) to model a large-scale explosion in a vented enclosure (10 × 8.75 × 6.25 m), revealing the issue of external explosion modeling. In order to reproduce the experimental results, the combustion rate needed to be increased by approximately a factor of 2 at the area outside the vent. A similar approach was followed by Makarov et al. [23] in vented explosion through relief pipe and by Tolias et al. [24] in hydrogen vented deflagration in a large scale chamber (FM-Global experiments). Bauwens et al. [25] and Keenan et al. [26] simulated FM-Global experiments too, focusing on the modeling of Rayleigh-Taylor instability for better reproduction of external explosion. The instability was modelled through an additional transport equation which takes into account the generation and the suppression rate of the instability.

In the current work, nine hydrogen vented deflagration experimental cases are simulated using the CFD method. The primary objective is the accurate prediction of the over-pressure until the time when external explosion occurs. Thus, we do not deal with modelling of flame-acoustic interactions that may enhance the combustion rate at later stages. A newly developed model is presented and evaluated against the experiments.

Description and analysis of experiments

In order to evaluate the new deflagration model, a hydrogen deflagration experiment performed by KIT in a medium scale enclosure [27,28] and vented deflagration experiments performed by FM-Global [25,29] in a large scale chamber are chosen. The choice was based on the similarity of geometries. Both facilities have very close dimensionless vent area providing the similarity of pressure records independent of 4 times different scales. The description of the experiments and discussion on their results are presented in the following sections.

KIT experiments

In KIT experiment [27,28], the facility consists of an empty enclosure of 0.98 × 0.96 × 1.0 m (length × width × height) with a vent 0.5 × 0.5 m in size located in the center of a front wall. The enclosure is positioned inside a room of 5.5 × 8.5 × 3.4 m at the distance of 1.94 m from the right wall, 2.6 m from the left wall, 1.95 m from the back-wall and 0.62 m from the ground. Pressure measurement devices were positioned both inside and outside the enclosure. A sketch of the test enclosure and instrumentation is presented in Fig. 1.

The comparison with the experiment is made in the current work using sensor P03 located inside the enclosure at the centre of the floor and sensor P09 located outside, at the distance of 0.24 m from the vent centre. Outside the enclosure the devices were placed upon a structure consisting of thin beams which was included in the geometrical model in the simulations. The beginning of the structure was placed at the distance of 0.15 m from the vent. The mixture is ignited at the center of the opposite to the vent wall (back-wall ignition). Before ignition, the thin plastic film which was used to hold the mixture was cut in order to minimize its effects. Visualization of the burned and the unburned mixture was made, using the BOS-schlieren technique. Thus, the flame front shape can be compared with the simulation results.

Experiments with homogeneous 7, 10, 12, 15 and 18% hydrogen-air mixture were performed. The experiment with 18% concentration is simulated in the current study. To evaluate the repeatability of the experiments, the 18% case was repeated four times under the experimental codes HIWP3-028, HIWP3-029, HIWP3-030, HIWP3-031.

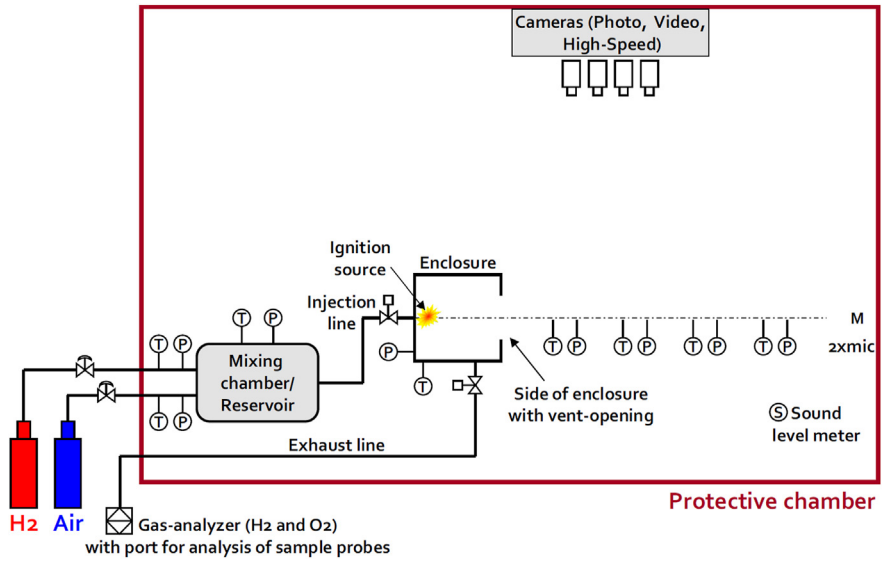

Fig. 1 – KIT experiments: Sketch of the test enclosure and instrumentation for experiments inside the Test-Room ($V = 160\ m^3$): T – thermocouples; P – pressure sensors; M – microphones.

Fig. 2 presents the overpressure inside and outside the enclosure for every repetition of the experiment. We observe that common features exist in all tests. Initially, the internal overpressure rises in exponential-like rate until it reaches the value of approximately 6 kPa. Then, the pressure increase stops and pressure remains constant for a short period of time. This steady period is interrupted by the sudden increase of the overpressure outside the vent. As a result, internal pressure starts to rise again but, this time, with much higher rate. We observe that in all experiments, the peak of the internal pressure occurs after the peak of the external one. These results indicate that the external explosion phenomenon, i.e. combustion of the unburned mixture that has been displaced out of the enclosure during the initial stages of the explosion, is responsible for the sudden increase of both internal and external pressures.

The external explosion effect can be confirmed by the examination of the flame images of the experiments [27,28]. In Fig. 3, internal and external overpressures are presented in separate diagrams along with experimental images from the HIWP3-030 and HIWP3-028 experiments (images from the other experiments are similar). In these images, both the flame front and the hydrogen-air mixture which is displaced out of the enclosure are visible. We observe that the time when the internal pressure stabilizes coincide with the time when the flame front passes through the vent. The sharp increase of the external pressure occurs after a while, when a significant volume of burned mixture has exited the enclosure.

Comparing the results from the different repetitions of the experiments, we observe that the measurements of the three last out of four experiments are very close to each other. In these three cases, maximum overpressure inside and outside the enclosure are approximately equal to 12 kPa and 8 kPa respectively. The experiment HIWP3-028 is the only one that deviated from the rest with the maximum internal pressure being approximately equal to 10 kPa and maximum external one approximately equal to 5 kPa. However, qualitatively, experiment HIWP3-028 is very similar to the other cases with pressure signals exhibiting similar trends and experimental images exhibiting similar flame shapes. The reason for the quantitative deviation of this experiment is unknown. The flame speed inside the enclosure is lower leading to lower pressure rise rate as the flame travels inside the enclosure and consequently leading to lower overpressure peaks inside and outside the enclosure.

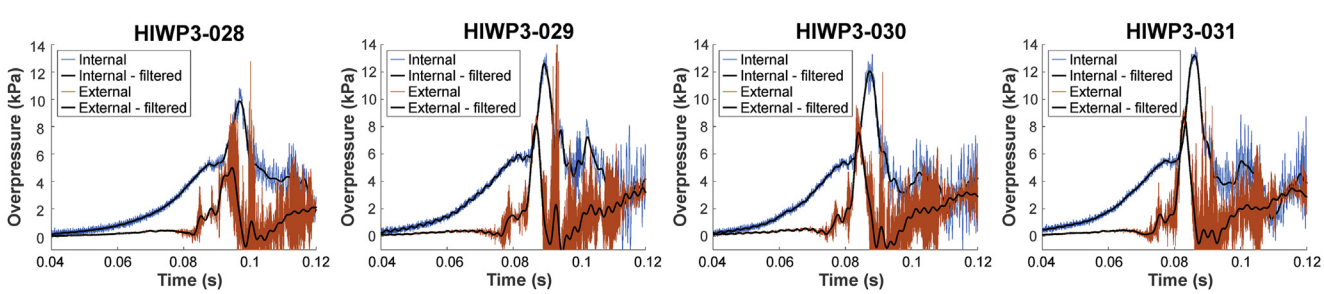

Fig. 2 – KIT experiments: Filtered (400 Hz) and unfiltered overpressure inside and outside the enclosure depicting the external explosion effect in every repetition of the 18% hydrogen-air mixture experiment.

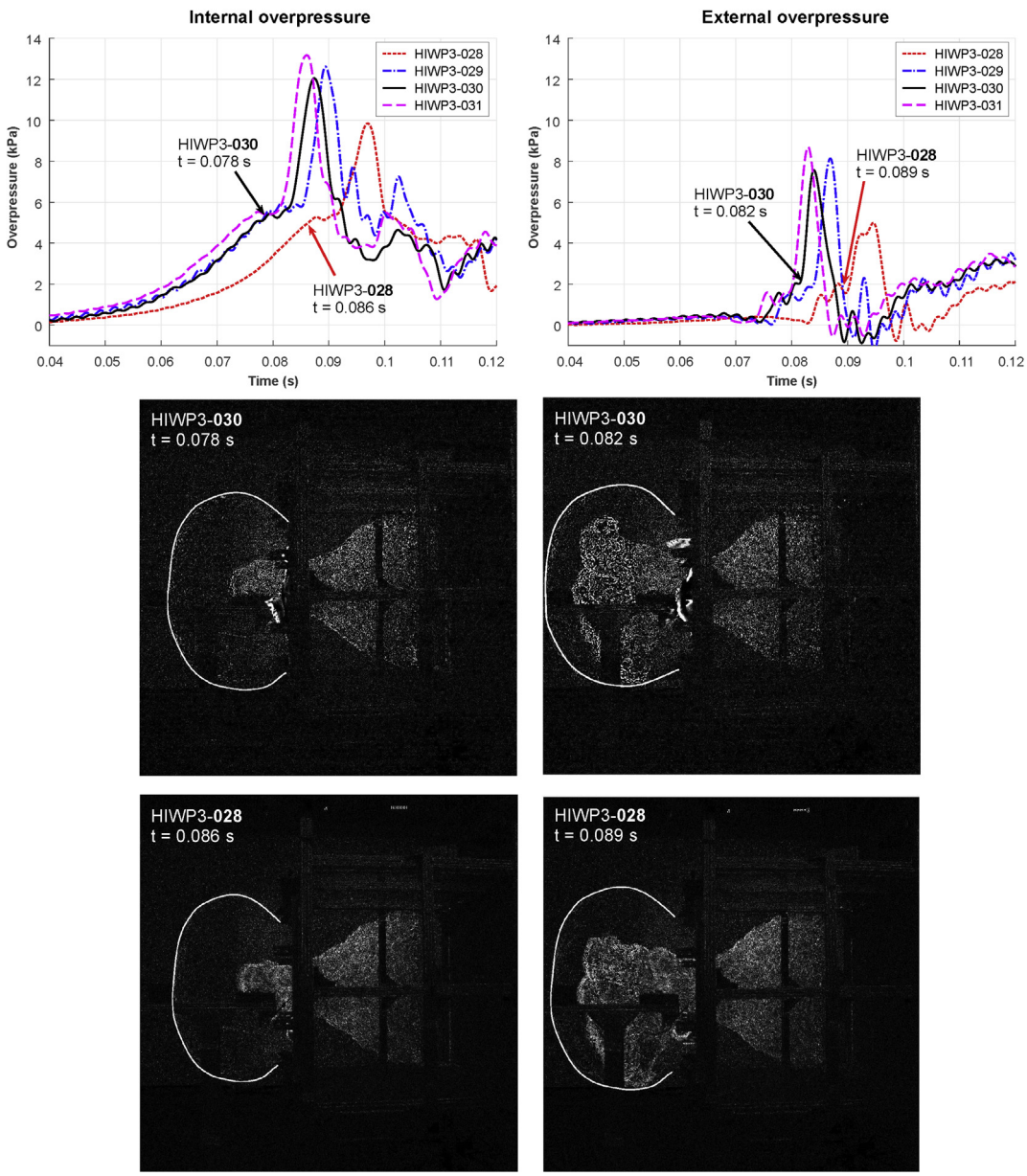

Fig. 3 – KIT experiments. Top: Comparison of the filtered overpressures inside (left) and outside (right) the enclosure. Bottom: Images of the flame along with the outline of the cloud outside the enclosure.

FM-global experiments

A series of hydrogen vented deflagration experiments with various concentrations, vent openings and ignition locations were performed by FM-Global [25,29,30]. The experimental facility consists of a square-floored enclosure of $4.6 \times 4.6 \times 3.0$ m (length × width × height) with a square vent. Eight cases are simulated in the current study with empty enclosure and vent area equal to 5.4 m^2.

In the first two cases, homogeneous 18% hydrogen-air mixture exists inside the enclosure which is ignited either near the back-wall or at the center of the enclosure. These experiments were initially conducted as it is descripted in Refs. [25,31,32]. In 2013, these experiments were repeated using a new chamber configuration [33]. In the new configuration, the frame structure outside the chamber and the flame time of arrival thermocouples was removed, the ignition source was changed and the vent cover was perforated prior to the test. The last change seems to be the most important one because it causes the vent cover to be removed completely at very low overpressure (less than 0.5 kPa [29]). On the other hand, in the initial configuration, the vent cover seems to be stretched significantly and then to rupture [33] due to the presence of the instrumentation outside the vent. This, among others, is expected to influence the shape of the hydrogen-air mixture outside the vent. Consequently, the effect of the vent cover on the results is much smaller in the new configuration.

Very small amounts of initial turbulence existed in these experiments due to the operation of fans that were used to

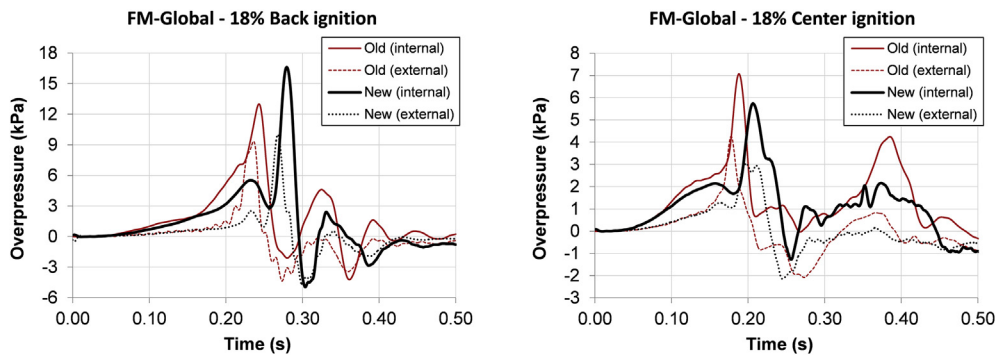

Fig. 4 – FM-Global experiments, 18% v/v hydrogen concentration [25,33]. Comparison of the overpressures time series for the old and new configuration for back (left) and center (right) ignition cases.

create the homogeneous hydrogen-air mixture. In the old configuration, initial turbulent intensity was estimated by using bi-directional velocity probe and found to be small (u' approximately equal to 0.1 m/s) [25]. The same ignition delay (time between the shutdown of the fans and the ignition of the mixture) was used in the new configuration as in the old one and thus the same turbulence levels are expected to exist.

In Fig. 4 the experimental results of these two cases are presented for the old and the new configuration. The internal pressure was measured at four positions within the enclosure with the filtered pressure time-series being the same for all sensors that were located away from the vent. The external pressure was measured on the concrete pad (at the same height as the bottom of the vent), 1.14 m away from the vent opening. We observe that the results from the different configurations have significant qualitative and quantitative differences. Qualitatively, in the new configuration a pressure drop occurs before the sudden increase of pressure at approximately 0.25 and 0.17 s for the back and center ignition case respectively. Moreover, in the center ignition case, the distinct second peak which is due to flame-acoustics interaction is almost vanished in the new configuration. Quantitatively, the pressure peak in the new configuration is bigger by approximately 28% in the back ignition case and smaller by approximately 19% in the center ignition case.

Consequently, the importance of minimizing the effect of the vent cover is depicted. Vent cover is difficult to be modelled with accuracy and thus experiments which minimize its impact on the results are necessary in order to increase the reliability of model validation.

Comparing with the KIT case, we observe that the external explosion effect appears in FM-Global experiments too. The overpressure inside the enclosure increases rapidly reaching the maximum value after the sudden increase of the external one. This is obvious in both old and new configuration. On the other hand, the overpressure drop in internal pressure that is observed before the external explosion does not occur in KIT experiment in which the pressure remains constant. A possible reason for this difference could be the different vent coefficient ($K = V^{2/3}/A_v$ where V the volume of the enclosure and A_v the area of the vent) of the two experiments. In KIT the vent coefficient is equal to 4 whereas in FM-Global is approximately equal to 3. Smaller vent coefficient means higher vent area and thus increased outflow which compensates the pressure increase. Regarding the maximum overpressure in back-wall experiments, its value between KIT and FM-Global is close due to the similar vent coefficient values.

In the other six experimental cases which are examined in the present study, the mixture composition is equal to 15% [29]. The vent cover was perforated prior to the test in these experiments too and a deployment pressure less than 0.5 kPa was measured. Three different initial level of turbulence (u' equal to 0.12, 0.40 and 0.50 m/s) and two ignition cases (back and center) are examined. The initial turbulence was created by mixing fans and determined by varying the ignition delay. The values of u' were estimated by using sonic anemometer to

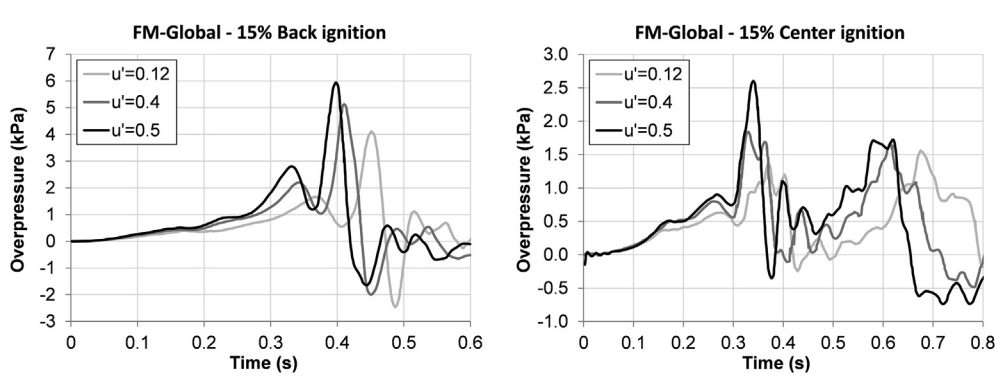

Fig. 5 – FM-Global experiments, 15% v/v hydrogen concentration [29]. Internal pressure time series for different levels of initial turbulence for back (left) and center (right) ignition cases.

measure the instantaneous flow velocity at nine points inside the chamber. More details can be found in Ref. [29]. The internal overpressure time series are presented in Fig. 5. Similarly to the 18% cases (Fig. 4), a pressure drop occurs before the sudden increase of pressure at approximately 0.35 and 0.28 s for the back and center ignition case respectively. We observe also that the increased levels of initial turbulence lead to greater overpressure peaks.

There are some significant differences between back and center ignition cases. The dependence of overpressure peaks on u' is not similar. The increase of u' from 0.4 to 0.5 m/s lead to 16% increase of the maximum overpressure in the back ignition case and to 41% increase in the center ignition case. The pressure signals for the center ignition case indicate a more complex flow with more local maximum to occur compared to the back ignition case, especially in the cases of u' equal to 0.12 and 0.4 m/s. The vent cover opening may be responsible for these differences. Vent opening may have a greater influence in the center ignition case because the flame arrives earlier at the vent compared to the back ignition case. Vent opening seems to be evident in the overpressure time series in the center ignition case, as overpressure remains constant when the value of 0.5 kPa is reached (at about 0.16 s).

Numerical modelling

Governing equations - combustion model

For the CFD simulations the ADREA_HF code was used [34,35]. The model solves the ensemble-averaged continuity equation, the Navier-Stokes equations, the energy equation (conservation equation of total enthalpy) and the conservation equation of hydrogen, oxygen and water mass fraction (one step irreversible reaction equation for hydrogen combustion is adopted). The assumption of thermodynamic equilibrium and of ideal gas equation of state is used for the multi-component mixture. Turbulence is modelled using the URANS (Unsteady Reynolds averaged Navier-Stokes equations) approach and the Kato and Launder modification of the k-ε model [36].

The newly combustion model that we recently presented in Ref. [37] is used and is described briefly in next. The reaction rate for hydrogen which appears in its conservation equation is modelled based on the turbulent burning velocity (S_T) concept [38]:

$$\overline{R}_f = \rho_u S_T \left| \nabla q_f \right| \tag{1}$$

where q_f is the fuel mass fraction and ρ_u is the density in the unburned region which is estimated using the assumption of adiabatic compression. The turbulent burning velocity is modelled considering all the main factors that affect the total reaction rate: laminar burning velocity, turbulence existing in front of the flame front, hydrodynamic instability and non-equidiffusive effects. It is estimated from the relations:

$$S'_T = S_L + u'\left(1 + Da^{-2}\right)^{-1/4} \tag{2}$$

$$S_T = \Xi_k \cdot \Xi_{lp} \cdot \Xi_{T_0} \cdot S'_T \tag{3}$$

In these relations, S_T is the turbulent burning velocity, S'_T the intermediate turbulent burning velocity which accounts only for the turbulence that exists in front of the flame front (Schmid's relation [39]), S_L the laminar burning velocity (function of pressure), Da the Damköhler number, u' is the rms (root mean square) of velocity fluctuations and Ξ are factors that account for the various mechanisms which accelerate the combustion process. The above formulation of the combustion model considers that the modelled phenomena are independent from each other and thus to wrinkle the flame at different scales [37]. We should note that as u' goes to zero, S_T goes to $S_L^{sgs} = \Xi_k \cdot \Xi_{lp} \cdot \Xi_{T_0} \cdot S_L$

Ξ_k accounts for hydrodynamic instability and self-turbulized flame front. The relation $\Xi_k = 1 + (\psi \cdot \Xi_k^{max} - 1) \cdot [1 - \exp(-R/R_0)]$ is adopted [40,41] where R is the distance from the ignition point, R_0 the transient distance which is required for the phenomenon to be fully developed, Ξ_k^{max} the maximum value of the wringing factor and ψ model parameter (between 0 and 1) representing the level at which the maximum theoretically value is reached. Ξ_k^{max} is estimated by the theoretically derived formula $\Xi_k^{max} = (E-1)/\sqrt{3}$ [42] where E is the expansion factor.

Ξ_{lp} accounts for non-equidiffusive effects. In lean hydrogen mixtures, non-equidiffusive effects make the flame unstable wrinkling its surface and increasing the combustion rate [43]. This instability is also known as diffusional-thermal instability and it develops from the very early stages of the explosion [44]. The value of the Ξ_{lp} factor is chosen equal to one on the ignition point and equal to a constant maximum value at a given radius where the instability has been fully established. This radius is set equal to 0.1 m based on the experimental data of Kim et al. (2015) [44]. The maximum value of Ξ_{lp} can be estimated [45] based on the leading point concept of Zimont and Lipatnikov [46]. This value depends on mixture concentration and for 18% hydrogen-air mixtures is approximately equal to 2. This is in agreement with the value of 1.96 which is derived from the experimental work of Bauwens et al. (2011) [25]. The value of 1.96 was used in this study for both 15 and 18% mixture concentration. Modelling diffusional-thermal instability by using a single constant factor is supported by the work of Bauwens et al. [25,29].

Compared to the model developed in Ref. [37], the coefficient Ξ_{T_0} has been added in this study in order to take into account the effect of initial turbulence. It is a constant value which is applied everywhere in the domain. Its value is greater than 1 only for the cases with significant levels of initial turbulence, i.e. the FM-Global 15% concentration cases with u' equal to 0.4 and 0.5 m/s. Bauwens et al. (2014) [29] suggests this modelling practice based on their observations of the initial turbulence effect on the flame speed. They modified their analytical model for vent sizing and reasonable agreement with the experiment was found. In CFD simulation, the effect of initial turbulence could be accounted for by using proper initial values of k and ε. However, ε is not known in the experiments. Simulations with various initial values of ε were conducted and the effect on the results was found to be significant. Consequently, we use very low initial values of k and ε (equal to 1.0e-06 m^2/s^2 and 1.0e-06 m^2/s^3 respectively for all cases) and the effect of initial turbulence is modelled with the

Table 1 – Examined cases and modelling parameters.

Examined cases					Modelling parameters				
No	Experiment	Conc. (v/v)	Ignition	Initial u'(m/s)	S_{L0}(m/s)	E	R_0 (m)	ψ	Ξ_{T_0}
1	KIT	18%	Back	–	0.64	5.20	1.0	1.0	1.00
2	FM-Global	18%	Back	~0.1	0.64	5.20	0.8	0.9	1.00
3	FM-Global	18%	Center	~0.1	0.64	5.20	0.8	0.9	1.00
4	FM-Global	15%	Back	0.12	0.38	4.68	0.3	1.0	1.00
5	FM-Global	15%	Back	0.40	0.38	4.68	0.3	1.0	1.12
6	FM-Global	15%	Back	0.50	0.38	4.68	0.3	1.0	1.17
7	FM-Global	15%	Center	0.12	0.38	4.68	0.3	1.0	1.00
8	FM-Global	15%	Center	0.40	0.38	4.68	0.3	1.0	1.12
9	FM-Global	15%	Center	0.50	0.38	4.68	0.3	1.0	1.17

addition coefficient. In this way, the validity of the approach suggested in Refs. [29] is also evaluated in CFD modelling.

The values of the main parameters of the model are presented in Table 1 for each examined case. Laminar flame speed at initial pressure, S_{L0}, and expansion coefficient, E, depend only on mixture composition. The parameters for the estimation of the Ξ_k coefficient, R_0 and ψ, were chosen based on guidelines [41,47] and in order to match the experimental overpressure at the initial stages of the back ignition deflagration cases. This means that for a given experimental set-up the same parameters are used in the back and center ignition case. Regarding R_0, guidelines [47] suggest maximum value equal to 1.2 m for hydrogen mixtures. This value is proposed to be decreased for lean mixtures where the transition to turbulence in the flame front is expected to occur quicker. Lower values are expected also if initial turbulence exists. As a result, in our cases, R_0 equal to 1.0 m is used in KIT experiment where near zero values of initial turbulence exists, 0.8 m in 18% FM-Global cases where small initial turbulence exists and 0.3 m in 15% FM-Global cases where higher values exist. The sensitivity of the results to the R_0 value was tested in some cases and is discussed in the results section. The value of ψ is set equal to 1.0 except from the two FM-Global cases with 18% hydrogen concentration for which the value of 0.9 is chosen.

Numerical details

In the simulations of the KIT experiment, computational domain (Fig. 6) includes the vented enclosure and the interior of the room in which the enclosure is. In solid boundaries no-slip boundary conditions were applied with standard wall functions. Three computational meshes were examined, consisting of approximately 714,000 cells, 1,539,000 cells and 2,726,000 cells. Cell size inside the enclosure was equal to 0.03, 0.02 and 0.015 m respectively. These cell sizes were also used outside the enclosure until the distance of 0.8 m from the vent in order to maintain the uniform mesh area in the region where external explosion occurs. Outside the uniform region, an expansion ration of 1.08 was applied. The dependency of the results from the computational mesh was small [37] and thus only the results from the fine mesh are presented in section Results and discussion.

In the simulations of the FM-Global experiments, symmetry in the y axis was assumed. The domain covers the area from −10 m to 40 m in the horizontal direction and from 0 m to 30 m in the lateral and vertical direction. The center of the room was placed at the origin of the coordinate system. Symmetry boundary conditions were applied in the south plane of the domain and non-reflecting boundary conditions in all other open boundary planes. Two computational meshes were examined, a coarse one with approximately 395,560, cells and a fine one with 893,412 cell. The control volumes size was approximately uniform inside the enclosure and outside until the distance of 4.2 m from the vent and equal to 0.15 and 0.10 m for the coarse and fine mesh respectively. The maximum expansion ratio outside that uniform region was equal to 1.1. The results from the mesh independency study are presented in the ssection Results and discussion.

ADREA-HF uses the finite volume method on a staggered Cartesian mesh. A modified SIMPLER algorithm is used in order to decouple pressure from Navier-Stokes equations [48]. Convective terms are discretized using the bounded second order upwind scheme [49]. Temporal discretization is made using the fully implicit first order accurate scheme. Time step

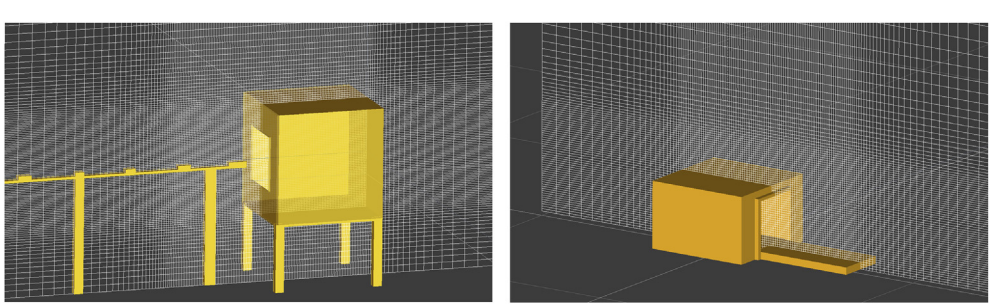

Fig. 6 – Geometry and part of the domain for the KIT (left) and the FM-Global (right) cases. The fine mesh in XZ plane passing through the center of the enclosures (symmetry plane in the FM-Global case) is also shown.

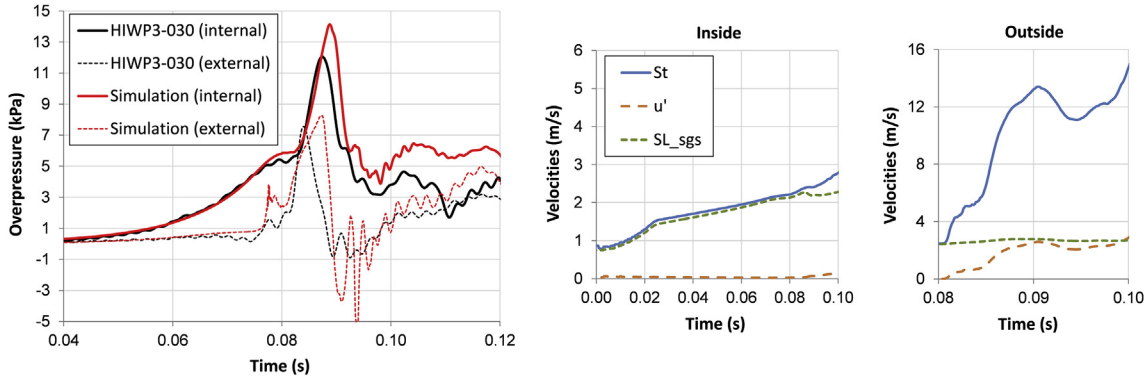

Fig. 7 – KIT case. Left: Measured and predicted overpressures inside and outside the enclosure. Right: Average predicted combustion related variables at the flame front existing inside or outside the enclosure.

is automatically calculated based on the Courant–Friedrichs–Lewy (CFL) number in order its maximum among all computational cells not to exceed the value of 0.2. Smaller CFL values were examined and negligible differences in the results were found. As initial conditions, a stagnant flow field is specified. Ignition is modelled by fixing the reaction rate in the computational cell of the ignition point, in order the initial amount of fuel to be burned at a determined interval estimated by the formula $0.5dx/(E \cdot S_L)$ where dx is the size of the cell at ignition point.

Results and discussion

Simulation of KIT experiments

A thorough analysis of the simulation results for this experiment can be found in Ref. [37]. In this study, only the main results are presented for comparison with the FM-Global simulations. In Fig. 7 (left) the simulation results for the overpressure inside (sensor P03) and outside (sensor P09) the enclosure are compared against experimental results. The measurements of the HIWP3-030 test are used for the comparison. Overall, CFD predictions are in very good agreement with the experimental measurements both inside and outside the enclosure. At the initial stage, when the flame is inside the enclosure the agreement is excellent. As the flame exits the vent (at 0.077 s) the pressure inside the enclosure remains approximately constant around 5.8 kPa, similarly to the experiment. The times when internal and external overpressures suddenly increase due to external explosion are also predicted very accurately. The rate of pressure rise inside the enclosure is also captured very well. The maximum overpressure, however, is overpredicted by approximately 17%.

In Fig. 7 (right) the predictions of the averaged combustion related parameters, i.e. turbulent burning velocity S_T , rms velocity fluctuations u' and $S_L^{sgs} = \Xi_k \cdot \Xi_{lp} \cdot \Xi_{T_0} \cdot S_L$ are presented for the flame existing inside or outside the enclosure. The average is taken over the flame front which exists only inside and only outside the enclosure. Inside the enclosure, u' values are negligible and thus no turbulence in front of the flame front is predicted on average. Consequently, flame instabilities drive the increase of the flame speed and S_T almost coincide with S_L^{sgs} in most moments. Outside the enclosure, u'

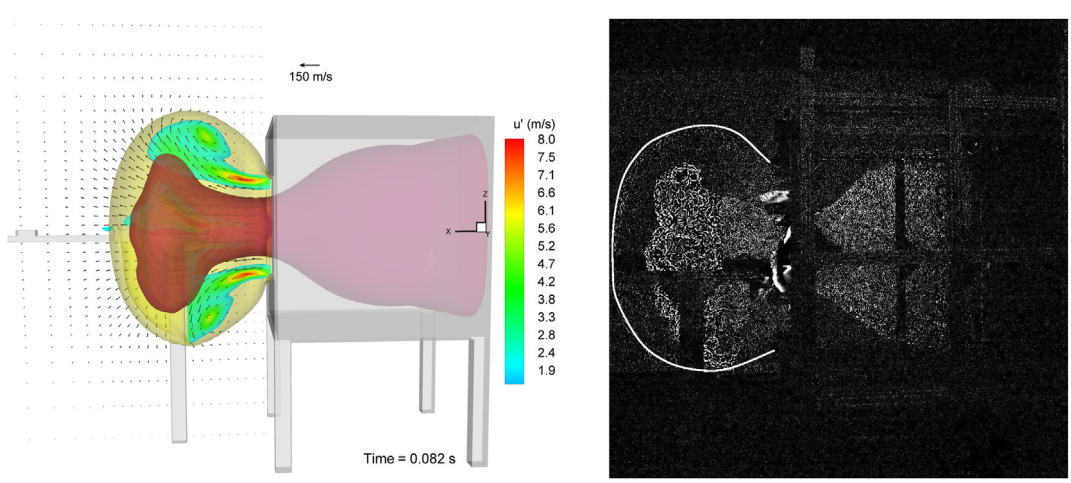

Fig. 8 – KIT case. Left: Flame front surface (dark red) and hydrogen-air mixture displaced outside the enclosure (yellow) along with u' contours (cutoff below 1.5 m/s) and velocity vectors at the onset of external pressure sudden increase (0.082 s). Right: Image from the experiment at the same instant. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

increases due to the increased levels of turbulence. As a result, S_T increases significantly deviating from S_L^{sgs}.

In Fig. 8 (left) the flame front surface and the hydrogen-air mixture displaced outside the enclosure are shown at 0.082 s. Contours of u' are also presented at the central plane along with velocity vectors. We observe that turbulence has been developed mainly in an area lateral of the vent edges. The presented instant corresponds to the beginning of the external pressure sudden increase. As a result, pressure increase coincides with the entrance of flame in the high turbulence area. In the same figure (on the right) the experimental image at the same time is presented showing both the flame front surface and the hydrogen-air mixture. The agreement between simulation and experiment regarding the shape and position of the surfaces is very satisfactory. Finally, axial flame position and axial flow position (position of the hydrogen-air mixture escaping the vent) were measured during the experiments. The comparison of the model results with the measurements exhibits a satisfactory agreement as it is presented in Ref. [37].

Simulations of FM-Global experiments

In Fig. 9 (left) the predicted and measured overpressures are shown for the FM-Global back ignition and 18% v/v hydrogen concentration case. The maximum internal overpressure is underestimated by approximately 14% whereas the maximum external overpressure is predicted almost perfectly by both meshes. The only significant difference between the examined meshes is that in the fine mesh case the overpressure remains approximately steady at the period 0.235–0.250 s whereas in the coarse mesh case the pressure continuously increases. However, the noticeable pressure drop in the experiment at the same period of time is not reproduced in both meshes. In Fig. 9 (right) the predictions of combustion related parameters for the flame inside or outside the enclosure are presented. We observe that, similarly to the KIT case (Section Simulation of KIT experiments), the effect of turbulence existing in front of the flame front on combustion is minor inside the enclosure whereas it is the dominant one outside the vent at the time when external explosion occurs.

In Fig. 10, u' contours along with the flame front are presented for the same case. The inner white line represents the flame front whereas the outer white line defines the area that is occupied by the hydrogen which has been pushed outside the enclosure. The first picture corresponds to the time when the internal overpressure has been stabilized (external pressure equal to 2.84 kPa at that time) whereas the second one at the onset of internal pressure sudden increase. The external pressure increase has begun in the second picture and it is equal to 4.8 kPa at that specific time. We observe that, similarly to the KIT case, the high turbulence area that has been developed outside the enclosure is responsible for the flame acceleration and the violence of the external explosion. The high turbulence

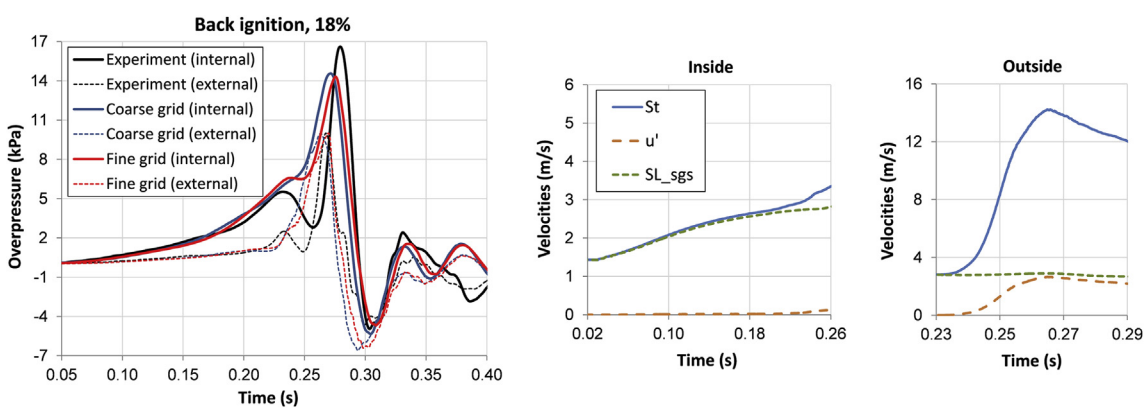

Fig. 9 – FM-Global back ignition case, 18% v/v hydrogen concentration: Overpressure inside and outside the enclosure (left) and average predicted combustion related variables at the flame front inside or outside the enclosure for the fine mesh simulation (right).

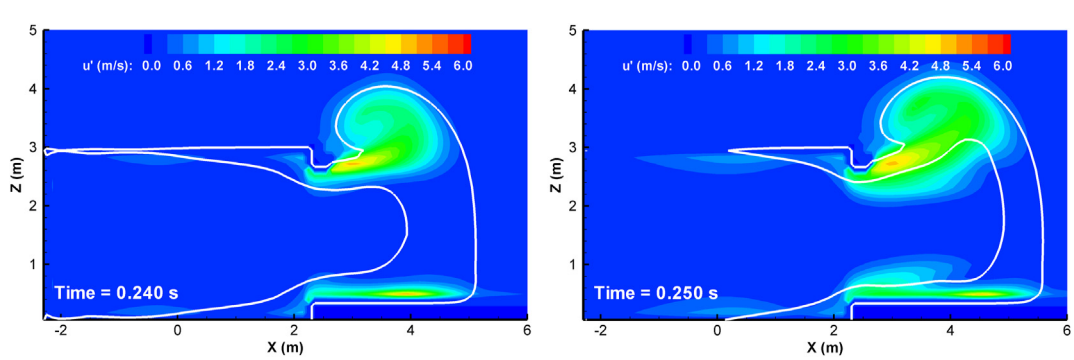

Fig. 10 – FM-Global back ignition case, 18% v/v hydrogen concentration: u' contours along with the flame front (inner white curve) and the outer limit of hydrogen cloud (outer white curve) at the symmetry plane.

area mainly exists away from the centreline of the opening at the areas of recirculation. This also holds for the XY-planes not shown here in which u' contours have similar shape.

In Fig. 11 (left) the results from the center ignition case and 18% v/v hydrogen concentration are presented. We observe that the two mesh sizes give almost identical results. Comparing with the experiment, the maximum internal overpressure is overpredicted by approximately 32% which is caused by the overprediction of the external pressure. At the initial phase of deflagration, a time delay in the predicted pressure exists compared to the experiment. In Fig. 11 (right) the predictions of combustion related parameters are presented. The model behaviour is similar to the one of the previous cases. Flame instabilities are responsible for the flame acceleration when the flame is inside the enclosure whereas turbulence is responsible for the flame acceleration in the external explosion.

In Fig. 12, u' contours along with the flame front and the hydrogen cloud are presented for the same case. The time 0.19 s corresponds to the time when the pressure inside the enclosure has been stabilized whereas the time 0.20 s corresponds to the time when the pressure increase has begun. Similarly to the back ignition case, pressure increases rapidly when the flame enters the high turbulence area that has been formed around the vent.

Fig. 13 presents the overpressure time series for the 15% hydrogen concentration cases and back-wall ignition for various levels of initial turbulence. Mesh independency was also performed in these cases and mesh convergence was very satisfactory, similar to the one presented for the 18% cases. The results with the fine mesh are presented. We observe that the agreement with the experiment is very good in all cases. Maximum overpressure is captured almost perfectly with maximum relative error being equal to 6%. The qualitative features of the time series are also predicted very well. The local maximum at the time when the flame exits the vent (e.g. at 0.33 s for the $u' = 0.50$ m/s case) is reproduced by the simulations, as well as the local minimum which follows. The value of this minimum, however, is overpredicted in all cases.

In Fig. 14, the results for the center ignition case are presented. We observe that the model overpredicts the maximum overpressure in all cases. The overprediction is equal to 54, 61 and 30% for u' equal to 0.12, 0.40 and 0.50 m/s respectively. The deviations from the experiments in the center ignition cases are likely to be partially caused by the omission of a vent cover opening model. In center ignition case, the flame is closer to the vent at the time of the vent opening and thus the results can be influenced in a greater extent compared to the back ignition case. This is evident in the experimental overpressure time series, as we comment in the section FM-global experiments.

The sensitivity of the results to the value of R_0 was examined in the $u' = 0.5$ m/s cases and it was found to be relative small. The value of 0.6 m was tested. In both ignition cases, a

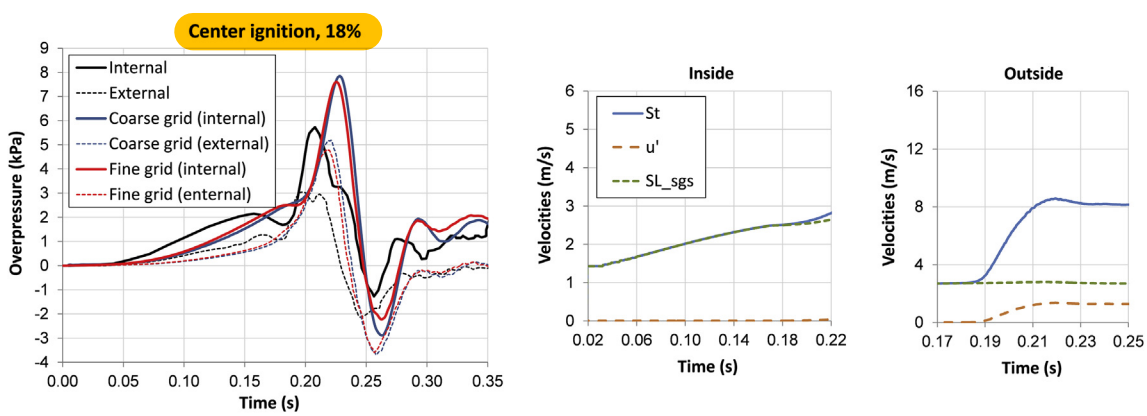

Fig. 11 – FM-Global center ignition case, 18% v/v hydrogen concentration: Overpressure inside and outside the enclosure (left) and average predicted combustion related variables at the flame front inside or outside the enclosure for the fine mesh simulation (right).

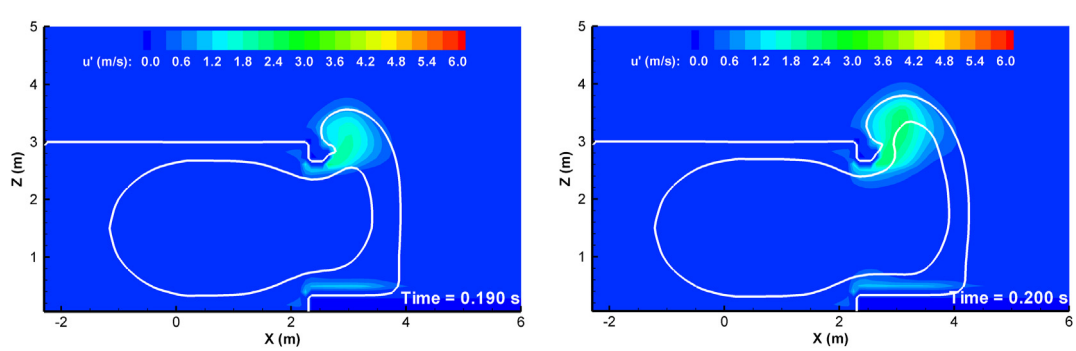

Fig. 12 – FM-Global center ignition case, 18% v/v hydrogen concentration: u' contours along with the flame front (inner white curve) and the outer limit of hydrogen cloud (outer white curve).

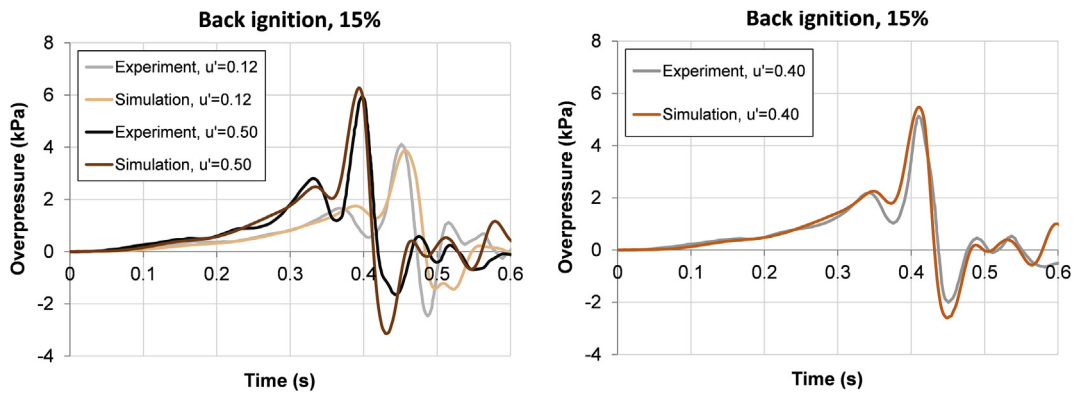

Fig. 13 – FM-Global back ignition case, 15% v/v hydrogen concentration: Overpressure time series for various values of initial turbulence - comparison between simulation and experimental results.

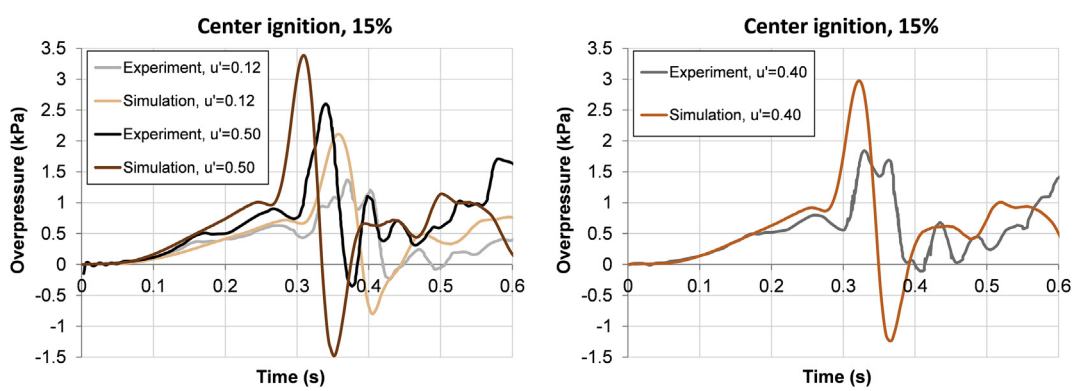

Fig. 14 – FM-Global center ignition case, 15% v/v hydrogen concentration: Overpressure time series for various values of initial turbulence - comparison between simulation and experimental results.

small time delay was created in the overpressure curves equal to 0.03 s without changing the overall shape of the pressure curves. The maximum overpressure was decreased, by 3% in the back wall ignition case and by 7% in the center ignition case. Such changes were expected and can be explained by the equation that is used to estimate the Ξ_k factor (see the section Governing equations - combustion model). It is interesting to note that this change led to better agreement with the experiment in the center ignition case. Consequently, the modelling of hydrodynamic instability may be partially responsible for the overestimation of the center ignition cases and it should be further evaluated in the future.

Conclusions

A newly developed CFD model for vented deflagration simulations was evaluated against nine experimental cases. The model exhibited very good performance for the back ignition cases. The maximum deviation from the experiments of the maximum overpressure was equal to 17% in those cases. The model reproduced successfully all the qualitative features of the overpressure curves with only exception the local minimum before external explosion in FM-Global 18% case. In center ignition cases, the model overpredicted the maximum overpressure up to 61%. However, study of the experimental results indicates that the opening of the vent cover may have a significant impact on the result in these cases. No model for vent opening was considered in this study. Some aspects of the employed model may also contribute to the overestimation of the center ignition cases, such as the modelling of the hydrodynamic instability. This could be a subject of future work for further improvement of the model.

Regarding the effect of initial turbulence, this was reproduced successfully by applying only a constant factor in the burning velocity. This is in agreement with the finding of experimentalists. The value of this constant is small being equal to 1.17 for u' equal to 0.50 m/s case.

Regarding the physics of the phenomenon, the results indicate that flame instabilities are primarily responsible for the acceleration of the flame front while it is inside the enclosure. Turbulence in front of the flame front mainly exists outside the vent at the recirculation areas leading to the significant increase of pressure when the flame enters these areas.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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